# 3D Structure Visualization Tools

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# Introduction

- 3D structure visualization tools are essential for understanding and analyzing the spatial arrangement of biological macromolecules such as proteins, nucleic acids, and complexes.
- These tools help researchers visualize, manipulate, and interpret molecular structures, which is crucial for various fields such as structural biology, bioinformatics, and drug design.
- <u>3D Visualization tools:</u>

    a) NCBI: CN3D, iCN3D

    b) RasMol, PyMol, Jmol

    c) SPDB Viewer

    d) Chimera

# Cn3D

- https://www.ncbi.nlm.nih.gov/Structure/CN3D/cn3d.shtml
- Cn3D is a software tool for the visualization of three-dimensional structures of proteins, nucleic acids, and other macromolecules.
- Integration with NCBI's Entrez database.
- Visualization of structure-annotated sequence alignments.
- Various representation styles and coloring schemes. Color can be used to determine different properties like domain, temperature.
- The menu consists of:

→ File: input data and output data (.png format)

→ View: Controls the structure as whole is displayed. Can zoom, restore, reset, control frame and animate

→ Show/Hide: Lets you focus on a particular chain. It has sequence viewer which shows sequence of structure.

# RasMol

- ❑ RasMol is a molecular graphics program intended for the visualization of proteins, nucleic acids, and small molecules.
- ❑ It was originally developed by Roger Sayle in the early 1990s.
- ❑ Simple and fast visualization.
- ❑ Runs to wide range of OS(Windows, Macintosh, UNIX)
- ❑ Input file format: PDB, Mol2, MDL/Molecular Design Limited, Xmol format, mmCIF
- ❑ <u>Molecule can be shown as:</u>

→ Wireframe bonds, Cylinder stick bonds, alpha carbon trace, hydrogen bonding, dot surface representation

→ Alternate conformers and multiple NMR model may be colored

→ Different parts colored and labeled properly

- ❑ RasMol includes a scripting language, to perform many functions such as selecting certain protein chains, changing colors, etc. Jmol and Sirius software have incorporated this language into their commands.
- ❑ RasMol Manual for command lines to operate and visualize the structure

  http://www.openrasmol.org/doc/rasmol.html

# RasMol

- Install RasMol:

Syntax In Command line to open a molecule: load filepath\filename.format

Syntax to change background: background color

Syntax to visualize hydrophobic residues: select hydrophobic

Syntax to Represent the hydrogen bonding of the protein molecule's backbone: hbonds on

### Command Reference

RasMol allows the execution of interactive commands typed at the '`RasMol>`' prompt in the terminal window. Each command must be given on a separate line. Keywords are case insensitive and may be entered in either upper or lower case letters. All whitespace characters are ignored except to separate keywords and their arguments.

All commands may be prefixed by a parenthesized '`atom expression`' to temporarily select certain atoms just for the execution of that one command. After execution of the command, the previous selection is restored except for the commands '`select`' , '`restrict`' and '`script`'.

The commands/keywords currently recognised by RasMol are given below.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Backbone | Background | Bond | Bulgarian | Cartoon | Centre | Chinese | Clipboard |
| Colour | ColourMode | Connect | CPK | CPKnew | Defer | Define | Depth |
| Dots | Echo | English | Execute | Exit | French | HBonds | Help |
| Italian | Japanese | Label | Load | Map | Molecule | Monitor | NoToggle |
| Pause | Play | Print | Quit | Record | Refresh | Renumber | Reset |
| Restrict | Ribbons | Rotate | Save | Script | Select | Set | Show |
| Slab | Source | Spacefill | Spanish | SSBonds | Star | Stereo | Strands |
| Structure | Surface | Trace | Translate | UnBond | Wireframe | Write | Zap |
| Zoom | | | | | | | |

### Background

```
Syntax:   background <colour>
```

### Bond

```
Syntax:   bond <number>  <number> +
          bond <number>  <number> pick
          bond rotate {<boolean>}
```

# PyMol


COMPND 2 MOLECULE: DNA (5'-D(*CP*GP*CP*GP*AP*AP*TP*TP*(CBR)
COMPND 3 P*GP*CP*G)-3');
COMPND 4 CHAIN: A, B;
COMPND 5 ENGINEERED: YES
ObjectMolecule: Read crystal symmetry information.
CmdLoad: "" loaded as "6bna".
PyMOL>
No License File - For Evaluation Only (0 days remaining)

# SPDB Viewer

- Swiss Protein Data Bank Viewer is a molecular visualization and analysis software specifically designed for the exploration of protein structures.
- It allows to analyze several proteins simultaneously
- SPDBV allows users to visualize protein structures in 3D using various rendering styles such as wireframe, cartoon, and surface representations same as RasMol and Cn3D.
- It is an application that provides a user-friendly interface allowing to analyze several proteins at the same time.
- The proteins can be superimposed in order to deduce structural alignments and compare their active sites or any other relevant parts.
- Amino acid mutations, H-bonds, angles and distances between atoms are easy to obtain thanks to the intuitive graphic and menu interface.
- It has been developped since 1994 by Nicolas Guex. Swiss-PdbViewer was initially tightly linked to SWISS-MODEL, an automated homology modeling server developed within the Swiss Institute of Bioinformatics (SIB) at the Structural Bioinformatics Group at the Biozentrum in Basel.
- However, the SWISS-MODEL web interface evolved to a point where it is now possible to use it directly for advanced modelling.
- Maintaining a direct interface with Swiss-PdbViewer is too complex and no longer supported.